

LawChecker: Bridging the Legal Knowledge Gap with AI

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1 Introduction

In a world where legal systems are often opaque, fragmented, and inaccessible to many, especially migrants and marginalized populations, the need for inclusive legal technology has never been more urgent. LawChecker is our contribution to bridging this global justice gap—a multilingual, explainable AI assistant designed to help users identify legal relevance, understand their rights, and access localized legal information. Motivated by a shared vision of justice through technology, we are submitting this project to *Prototypes for Humanity 2025* in hopes of amplifying its impact and connecting with a global community of innovators. Our system integrates advanced AI techniques, including Retrieval-Augmented Generation and jurisdiction-aware semantic search, into an accessible, human-centered design. LawChecker demonstrates how responsible use of technologies such as LangChain and Hugging Face can empower individuals, increase legal literacy, and promote fairness across borders. This paper outlines the legal, social, and technical foundations of LawChecker, and our long-term goal to scale its capabilities globally. **All components and implementation details of this research have been completed and are publicly available in our GitHub repository. The repository includes the full source code, detailed reproduction instructions, and a demonstration video of the LawChecker system in action. Interested readers may access it at: <https://github.com/gregalexan/prototypesforhumanity>**

2 Background & Related Work

Traditional models of access to justice—often rooted in formal court procedures shaped by Eurocentric paradigms—tend to overlook the needs of marginalized communities [1]. These frameworks emphasize official legal processes and procedural fairness, yet in practice they often exclude groups like migrants who cannot easily navigate such systems. In fact, roughly two-thirds of the world’s population (about 5.1 billion people) lack meaningful access to justice, and this deficit is especially acute for migrants[2]. Migrants frequently face cultural and linguistic barriers, unfamiliar legal systems, economic hardships, and other structural obstacles that make seeking legal help daunting [1]. Under a court-centric status quo, many migrants and other disadvantaged individuals are effectively shut out from the protections of the law, contributing to a global justice gap.

Over the past decade, various digital tools — from static FAQ websites to early legal chatbots — have attempted to improve access to legal information. However, these tools have serious limitations. They often provide only generic, one-size-fits-all answers and lack coverage of specific national or local legal frameworks [3]. For example, a chatbot trained on general information may struggle to incorporate the nuances of different countries’ laws or languages, leaving users with shallow guidance that isn’t well tailored to their situation [4]. Moreover, early legal AI systems typically offer limited transparency about how

they reach conclusions. Users receive an answer but gain little insight into the legal reasoning behind it, making it hard to trust the advice [5].

Recent advances in generative AI promise more sophisticated help, but so far they have not been widely adopted in the justice sector — and for good reason. Generative AI chatbots like ChatGPT can produce impressively human-like legal answers, yet they are prone to critical errors. In some cases, AI systems have even fabricated legal citations or case law out of thin air, misleading users and professionals alike [6]. Such “hallucinations,” along with the potential for biased or unfair recommendations, pose serious risks in legal contexts [5]. Experts caution that without proper safeguards, these AI tools might reinforce existing biases or provide dangerously inaccurate advice, ultimately harming the very communities they aim to help [4]. These concerns over accuracy, bias, and accountability help explain why human-centered AI solutions for access to justice remain nascent. The opportunity is clear: responsible AI could help people recognize legal issues, understand their rights, and take informed action. But realizing that potential will require overcoming the current trust gap by ensuring AI systems are reliable, transparent in their reasoning, and aligned with local legal realities [3].

3 Interview

To better understand the real-world legal challenges faced by migrants in Greece and assess the practical value of LawChecker, we conducted an interview with a practicing legal professional Maria Flouraki (<https://flouraki.gr>). The conversation focused on the gaps in legal access experienced by asylum seekers and undocumented migrants, and how a tool like LawChecker could support both legal professionals and vulnerable individuals.

The interviewee is a licensed attorney based in Heraklion, Crete, who has operated a private law office since 2003. Since 2019, she has also been a registered member of the official lawyer registry for Greece’s Asylum Service, providing legal aid at the second-instance level to applicants for international protection (including asylum and subsidiary protection). Her legal practice includes both administrative cases—such as residence permits and asylum status decisions—and criminal cases, including those related to migrant smuggling and border infractions.

Q: What are the biggest obstacles migrants face when trying to exercise their legal rights?

A: The primary and most significant barrier is the lack of language knowledge. Most documents and legal procedures are conducted in Greek. Often, even the lawyers assisting migrants—whether through organizations or as volunteers—do not speak their native languages. In some cases, they may not even speak English. This creates a serious communication problem. Interpreters used in these procedures are often not professionals, which results in poor or even incorrect interpretation. This prevents migrants

from fully understanding their rights. Additionally, most refugees and migrants cannot afford to hire a lawyer, and free legal assistance is limited only to specific procedures—not all of them.

Q: Are there cases where migrants are unaware they’ve committed a legal offense?

A: Yes, absolutely. Some clients are not even aware that their actions might be illegal. One extreme example involves alleged migrant smugglers: A very young—or even underage—migrant boards a boat from Libya to reach Europe. At some point, he is forced to take the wheel for survival reasons—perhaps because he cannot afford the fare and this is his only chance to escape. When he arrives in Greece, he is prosecuted as a smuggler, without realizing that what he did constitutes a criminal offense.

Q: How do language and cultural differences affect legal proceedings?

A: These differences cause misunderstandings on both sides. Migrants often struggle to clearly present their situation to judges. In turn, judges may not adequately explain the charges or rights to the defendants. This issue can only be resolved through professional and high-quality interpretation.

Q: Are there other cultural obstacles beyond language?

A: Definitely. For example, a woman who has been raped and comes from a very conservative country may be afraid to speak up due to shame or guilt, rooted in cultural norms. The same applies to members of the LGBTQ+ community. There are also individuals suffering from post-traumatic stress or other mental health conditions, making it difficult for them to clearly explain why they left their home country. In such cases, trained interpreters and mediators are essential to help them overcome fear, talk freely, and describe the persecution they’ve experienced.

Q: Do refugees generally trust the legal system?

A: Many refugees do not trust state authorities because they have suffered abuse and persecution in their countries of origin. They fear speaking openly about what happened. It is crucial to convince them that the system here is not like the authoritarian regime they escaped from, and that their statements will not be shared with their home countries. They need to feel safe and free to talk.

Q: Do you believe a tool like a legal chatbot—which explains in simple terms when a problem has legal relevance—would be helpful?

A: Absolutely. It would provide an initial layer of information and guidance, helping people understand whether they need specialized legal advice.

Q: In what ways could AI support a lawyer’s work? Are there any risks involved?

- A:** AI can definitely assist a lawyer—for example, by helping search for relevant case law or legal texts. It can be a tool to support legal work, but not a replacement. It should never substitute the role of a lawyer, only enhance it.
- Q:** What do you consider the most critical feature of a legal support app for migrants?
- A:** First, multilingual support—it must be available in many languages, even less common ones spoken in the migrants’ countries of origin. Second, it should connect users with legal advisors specialized in migration and asylum law.
- Q:** What would help migrants develop more trust in the legal system?
- A:** Clear and simple explanations—understanding the role of the lawyer, their rights, and their responsibilities in plain language.
- Q:** Is there a lack of legal literacy among refugees and migrants? How can it be addressed?
- A:** Without a doubt, there is. Many have low levels of formal education or come from countries without democratic institutions. This makes it even harder for them to grasp legal processes. Strengthening their legal education is essential.

In addition to her legal experience, the interviewee holds a master’s degree in the management of refugee and migration flows in Greece and Europe (University of the Aegean, Department of Sociology). Based on this experience, they emphasized how digital tools like LawChecker could assist in helping migrants better understand their legal position, while also aiding lawyers in the quick retrieval of relevant legislation—especially in fast-paced, high-stakes asylum cases.

4 Legal Access Barriers

Despite the foundational role of law in democratic societies, large portions of the global population remain unable to access it meaningfully. According to the Victoria Law Foundation’s Public Understanding of Law Survey (PULS), a significant number of citizens—particularly young people, non-English speakers at home, and those in financial distress—fail to recognize the legal relevance of problems in their daily lives. This legal unawareness is strongly correlated with socioeconomic vulnerability [7]. Migrants, in particular, face intensified legal exclusion. The dominant Eurocentric and state-centric conceptions of access to justice, focused narrowly on courts and litigation, often overlook migrants’ substantive needs and local justice realities [1]. As a result, many do not even recognize that an injustice has occurred, let alone seek redress.

Globally, 5.1 billion people—two-thirds of the world’s population—lack access to justice. Of these, 253 million live in extreme legal exclusion, including

stateless individuals, modern slaves, and those in conflict zones [2]. In the U.S., only 37% of immigrants in deportation proceedings have legal representation, a disparity largely driven by financial hardship and the absence of public legal aid infrastructure in rural or conservative regions [8]. Cultural and linguistic barriers further deepen this divide: in Canada, highly educated immigrants reported severe difficulty navigating government legal websites and services due to inaccessible language and complexity. Many expressed that they simply “did not know what to do or where to go” when facing legal problems [9]. This disconnection fosters legal cynicism—the belief that legal institutions are illegitimate or ineffective—a concept validated in multiple sociological studies and shown to reduce the likelihood of seeking legal help even when available.

These findings emphasize that legal exclusion is not only a structural issue but also an epistemic and psychological barrier to justice. **Therefore, bridging this gap requires not just infrastructural improvements, but also innovative, user-centered tools that empower individuals to recognize, understand, and act upon their legal rights.**

5 Methodology

5.1 System Overview

LawChecker is an AI-powered assistant that enables citizens to ask questions about potential legal scenarios in natural language and receive localized, understandable guidance. The system aims to help individuals identify whether a situation may have legal relevance, provide contextual information from national legal frameworks, and explain the rationale behind each response. LawChecker is designed to be multilingual, locally adapted, and explainable—prioritizing accessibility, legal literacy, and trust in justice systems across diverse user populations.

5.2 System Architecture

The overall system architecture of *LawChecker* is composed of a chat-based user interface and a multi-layered backend designed for jurisdiction-aware legal reasoning. The frontend provides a conversational interface—accessible via web platforms—through which users can input questions about potential legal issues in natural language made with react.

The backend integrates a generative AI language model (LLM), such as OpenAI’s GPT-4, orchestrated via the LangChain framework for structured interaction using Python and FastAPI. Legal reasoning is grounded through a Retrieval-Augmented Generation (RAG) pipeline, which dynamically fetches relevant statutory texts to support accurate and explainable responses. To enable localization, we gathered and processed two primary sources of legal content: (1) the Charter of the United Nations, related international treaties and Human Rights, and (2) national legal texts from Greek law. These documents

were scraped from official repositories and converted into plain-text format.

Following preprocessing and cleaning, the legal corpus was embedded using sentence-transformer models and stored in a FAISS vector index for efficient similarity search. This allows the system to retrieve relevant legal passages at runtime based on semantic proximity to the user’s query. Prompt engineering layers the results into a structured context window, with embedded role instructions and disclaimers to prevent overreach or misinterpretation.

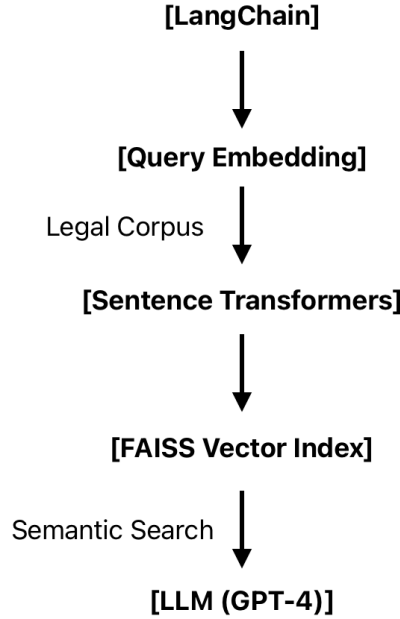


Figure 1: Workflow of the LawChecker chatbot answering a legal question.

All responses include a mandatory disclaimer clarifying that the output does not constitute legal advice, and results are supplemented with citations or article references when available. A logging and traceability module records user interactions and model outputs to ensure transparency and auditability. This architecture allows *LawChecker* to deliver responsive, jurisdiction-sensitive legal guidance while preserving user trust and institutional accountability.

5.3 Legal Reasoning Flow

Each user query undergoes a multi-step analysis pipeline:

1. Natural language understanding identifies whether the query likely concerns a legal matter (e.g., employment, tenancy, immigration).

2. If a legal domain is detected, the system retrieves relevant law articles or rules from the appropriate jurisdiction, if available.
3. The AI model generates a response that includes a simplified explanation, supporting references, and a disclaimer that clarifies the output is not formal legal advice.

This flow enables LawChecker to assist users without overstepping the role of a legal professional.

5.4 Localization and Adaptation

To ensure responses are legally accurate and socially relevant, LawChecker incorporates localization mechanisms:

- Law mappings are maintained for each supported jurisdiction to guide retrieval and interpretation.
- Language adaptation supports multiple languages, including plain language versions of complex terms.

Localization ensures that LawChecker remains useful across different legal and cultural contexts.

5.5 Safety Measures and Disclaimers

To mitigate risks and build user trust, LawChecker includes the following safeguards:

- Every response is accompanied by a disclaimer: “This is not legal advice. For formal legal support, consult a licensed attorney.”
- Sources or reference articles are cited when available, especially in retrieval-based outputs.
- When confidence is low or law is ambiguous, the system transparently communicates uncertainty (e.g., “This may depend on your location or specific case details.”)

These safeguards are essential to ensure ethical deployment of AI in sensitive legal contexts.

5.6 Prototype

The LawChecker prototype has been implemented as a web-based chatbot interface that integrates AI-powered legal reasoning with localized content. Below, we present screenshots from various components of the system, including the homepage, language selection, use case examples, and the system’s legal reasoning capabilities.

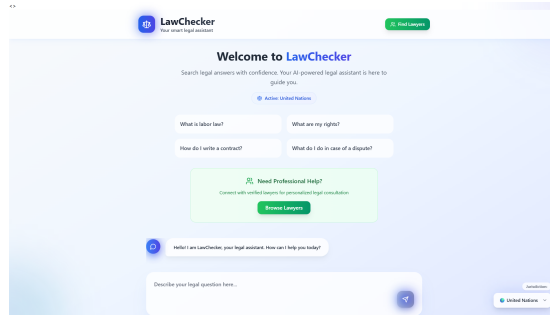


Figure 2: Homepage of the LawChecker web application.

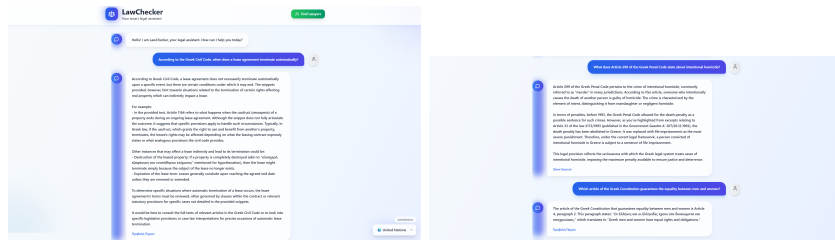


Figure 3: Greek-language interface displaying a user query and system response.

A video demonstration of the system in action is available on YouTube: <https://www.youtube.com/watch?v=F3woqWTIeno>

The full source code, installation guide, and reproduction instructions are available on GitHub: <https://github.com/gregalexan/prototypesforhumanity>

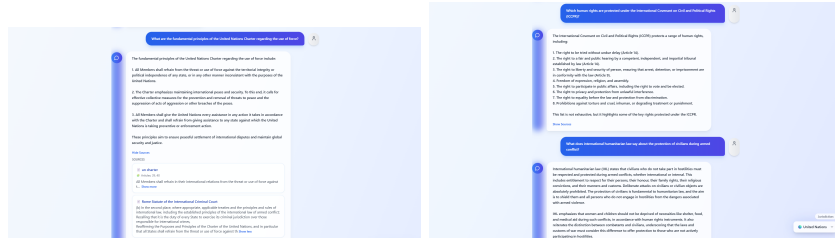


Figure 4: Examples of international law responses based on United Nations charters.

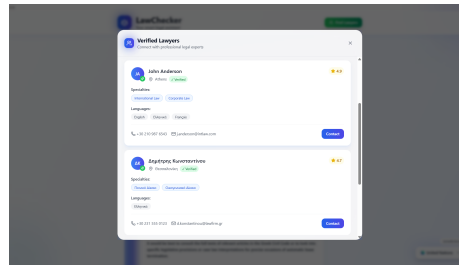


Figure 5: Prototype feature helping users locate legal aid resources.

6 Discussion

Despite the promise of generative AI, deploying LLMs in the legal domain remains fraught with challenges. One of the most pressing issues is the rapid pace at which laws evolve. Legal frameworks are constantly amended, repealed, or reinterpreted through court rulings, making it critical for any AI system operating in this space to stay up to date. In the context of Greek law, this challenge is particularly pronounced due to the lack of freely accessible, machine-readable legal databases. Platforms like *nomos.gr* provide structured legal content, but access typically requires paid credentials and manual navigation, making automated updates difficult without specialized scraping infrastructure and agreements for API-level access. Furthermore, LLMs—even when integrated into Retrieval-Augmented Generation pipelines—can hallucinate, misinterpret jurisdictional nuances, or omit critical disclaimers if not carefully instructed and monitored. LawChecker does not attempt to resolve these systemic issues by replacing lawyers or legal institutions. Instead, it is designed as an assistive layer—providing informational support to both laypeople and legal professionals. For non-experts, it can serve as a first step toward recognizing that a situation might have legal implications, empowering them to seek help. For lawyers, it can act as a triage and research assistant, helping them quickly identify relevant provisions and precedents. LawChecker’s role is not to make legal decisions, but to enhance legal literacy and reduce the asymmetry of knowledge between citizens and the legal system.

7 Conclusion and Future Work

LawChecker represents a promising step toward democratizing legal knowledge and addressing the global justice gap. By combining generative AI with local legal data and a user-centered design, the system empowers individuals—especially those from marginalized communities—to better understand and navigate their rights. Built using innovative frameworks such as LangChain for structured AI interaction and Hugging Face embeddings for semantic legal search, LawChecker demonstrates how cutting-edge technologies can be responsibly applied to sensitive domains like law. Looking ahead, our vision is to expand LawChecker beyond its current scope of Greek national law, United Nations Rights and international human rights frameworks to cover a broader spectrum of legal systems worldwide. This includes integrating jurisdiction-specific laws from countries across all continents, with a special focus on regions where migrants, minorities, and underserved populations face persistent legal exclusion. Our ultimate goal is to build a global infrastructure—comprising both a web platform and a mobile application—for legal literacy that is inclusive, transparent, and ethically aligned—ensuring that no one is left behind when it comes to accessing justice.

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